Intermediate Evaluation 1

Abstract (part that will be developed further)

The main goal of the research is to develop and evaluate AI-based models for demand forecasting in the fashion retail industry. The study aims to explore how machine learning and deep learning techniques can be used to predict product-level or category-level sales more accurately than traditional forecasting methods. By analyzing historical sales data along with influencing factors such as seasonality, pricing, and promotions, the research seeks to identify the most effective modeling approach for improving forecast accuracy.

Literature Overview

Demand Forecasting for Multichannel Fashion Retailers by Integrating Clustering and Machine Learning Algorithm

Using empirical data from UNIQLO (a major fast fashion retailer) spanning 12 years, the study compares the performance of these hybrid cluster-based models (KM-ELM and KM-SVR) against traditional, single-model benchmarks (simple ELM and SVR). The research analyzes sales data across both physical stores and non-store (virtual) channels. The findings demonstrate that the proposed hybrid models, specifically KM-ELM, significantly outperform standard models in prediction accuracy. However, the study also reveals that increasing the number of predictor variables—specifically by adding aggregated meteorological data—does not necessarily improve accuracy due to data relevance and dimensionality issues.

Key points:

1. **Challenges in Fast Fashion Forecasting** The fast fashion business model has shifted supply chains from push to pull production, requiring rapid responses to consumer needs. However, the nature of the industry makes forecasting difficult.
2. **The Efficacy of Clustering-Based Models** The authors argue that feeding all training data into a single model can lead to inaccuracies due to data heterogeneity. They propose using K-means clustering to group data with similar characteristics before applying regression models.
3. **Comparison of Machine Learning Algorithms** The study compares Extreme Learning Machines (ELM) and Support Vector Regression (SVR). While simple SVR outperformed simple ELM in isolation, the integrated KM-ELM model achieved the highest overall accuracy.
4. **The Impact of Variable Selection and Weather Data** The study tested two sets of predictor variables: a 10-variable set (historical sales and BIAS indicators) and a 24-variable set (adding temperature and rainfall data). Counterintuitively, the inclusion of weather data slightly decreased accuracy.

Deep Learning for Demand Forecasting in the Fashion and Apparel Retail Industry

This research paper addresses the significant challenges associated with demand forecasting in the fashion retail industry, specifically for new products that lack historical sales data. Traditional forecasting models, which rely heavily on historical times-series data, are insufficient for the fashion industry due to short product life cycles and high market volatility. To overcome this, the study proposes a novel "intelligent forecasting system" that integrates visual data (product images) with historical sales data.

The methodology involves extracting features from product images using a pre-trained Deep Learning model (Inception V3) and clustering historical products based on their sales profiles. A classification model is then trained to assign new products—based on their images—to these established sales clusters. Finally, the system predicts the weekly sales of the new item by identifying its "nearest neighbour" within the assigned cluster. The study utilizes data from a European fashion retailer comprising 290 items from 2015 and 2016. The results indicate that a Neural Network (NN) model performed best for classification, and the overall system showed promising results for short-term weekly sales prediction.

Key points:

1. **The "Cold Start" Problem in Fashion Forecasting** The authors highlight that the primary difficulty in fashion forecasting is the introduction of new items that have no sales history to analyze.
2. **Utilizing Unstructured Image Data** The study proposes that visual attributes (style, pattern, color) contained in product images can serve as a proxy for historical data.
3. **Clustering Methodology** The researchers used K-means clustering to group existing products based on their sales profiles. They determined that the optimal number of clusters for their dataset was two.
4. **Superiority of Neural Networks for Classification** Four classification models (Support Vector Machine, Random Forest, Neural Network, and Naïve Bayes) were tested to categorize images into the sales clusters.The Neural Network performed best.
5. **Forecasting via "Nearest Neighbour"** Once a new item is classified into a cluster, the specific sales forecast is derived by finding the most visually similar historical item within that cluster.

Deep Learning based Forecasting: a case study from the online fashion industry

The authors describe a custom Transformer-based architecture designed to address specific industry challenges, such as high catalog turnover (cold start problems), heavy-tailed demand distributions, and the "fixed inventory assumption," where stock cannot be replenished during a season.

The primary innovation described is the integration of a "monotonic demand layer" within the neural network. This layer ensures that the model learns a simplified, economically logical relationship between price discounts and demand (i.e., lower prices yield higher demand), which is crucial because the forecasts serve as inputs for a downstream pricing optimization system. The study demonstrates that this architecture outperforms traditional statistical methods (like DeepAR and LightGBM) and provides empirical evidence that Transformer-based forecasting models benefit from "scaling laws" similar to those observed in Natural Language Processing (NLP).

Key points:

1. **Unique Challenges in Online Fashion Forecasting** The authors identify specific constraints in the fashion industry that differentiate it from other retail sectors (like

grocery). The most significant is the "fixed inventory assumption," where stock is bought ahead of the season and cannot be reordered. Additionally, the data is characterized by extreme sparsity and frequent "cold starts" for new items.
2. **Distinction Between Sales and Demand** The model must predict *demand* (customer intent), not just *sales* (completed transactions). In fashion, sales are often capped by stock availability (e.g., specific sizes being out of stock). The authors use a pre-processing step to impute true demand based on the availability of different sizes.
3. **Transformer Architecture with Monotonic Demand Layer** The forecasting model utilizes an encoder-decoder Transformer architecture. A critical component is the "Monotonic Demand Layer," which models demand as a piece-wise linear function of the discount.This incorporates an inductive bias to prevent the model from learning spurious positive correlations between high prices and high demand (confounding).
4. **Strategic Training for "Near Future" Accuracy** The model separates the forecasting horizon into "near future" (first 5 weeks) and "far future" (up to 20 weeks). Training is weighted heavily toward the near future because that is the timeframe in which pricing decisions are actionable.
5. **Empirical Success and Scaling Laws** The Transformer model outperformed baseline models (Naive, LightGBM, DeepAR) in terms of Demand Error and RMSE. The authors also provide evidence that the model adheres to scaling laws, meaning performance continues to improve as the training dataset size increases.

A Demand Forecasting Model Leveraging Machine Learning to Decode Customer Preferences for New Fashion Products

This research article proposes a novel demand forecasting model for the fashion industry designed to address the challenges of unpredictable consumer preferences and the lack of historical data for new products. Traditional sales data is insufficient for predicting demand for new collections because it only reflects final purchases, missing out on "potential sales."

The study utilizes **fitting room data** from a retail store in India to build customer profiles based on garment features such as size, color, pattern, and price. The methodology employs a two-step machine learning approach:

1. **Unsupervised Learning (K-means Clustering):** To segment customer preferences into distinct clusters.
2. **Supervised Learning (Decision Tree Classification):** To classify new products into these established clusters to predict their potential demand percentage.

The findings suggest that this data-driven approach allows retailers to stock products that align more improved accuracy with customer preferences, thereby minimizing excess inventory and revenue loss.

Key points:

1. **The "Cold Start" Problem in Fashion Forecasting** The fashion industry struggles with forecasting because new collections lack historical sales data, and trends shift rapidly. Retailers often overstock or understock based on intuition rather than data.
2. **The Superiority of Fitting Room Data** The authors contend that fitting room data provides a more complete picture of consumer interest than sales data because it captures items customers considered but did not buy (missed opportunities).

3. **Feature Engineering and Importance** Using machine learning models (Decision Tree, Random Forest, and Lasso Regression), the study identified the most critical product features influencing customer choice.
4. **Clustering Methodology (K-Means)** The study used K-means clustering to group similar customer profiles. Using the silhouette score and elbow method, they determined the optimal number of clusters was six.
5. **Predicting Demand for New Products** Once clusters were weighted based on their volume in the fitting room, a decision tree classified new garments into these clusters to forecast their sales potential.

Retail Demand Forecasting: A Comparative Study for Multivariate Time Series

The authors argue that excluding external economic factors leads to less accurate predictions. To bridge this gap, the study proposes enriching time series data with macroeconomic variables, specifically the Consumer Price Index (CPI), the Index of Consumer Sentiment (ICS), and unemployment rates.

Using five years of sales data from Walmart locations in California, Texas, and Wisconsin, the researchers tested five different machine learning models: Lasso, Ridge regression, Light Gradient Boosting Machine (LGBM), Extreme Gradient Boosting Machine (XGBM), and Decision Tree. The study compares the performance of these models with and without the inclusion of macroeconomic data. The results confirm that integrating economic indicators generally improves forecasting accuracy, with the LightGBM model demonstrating the most significant performance gains.

Key points:

1. **The Limitation of Traditional Forecasting** The authors contend that current forecasting literature is insufficient because it fails to account for the broader economic environment that influences consumer behavior.
2. **Integration of Macroeconomic Variables** The core methodology involves merging specific economic indicators with standard sales data to create a "comprehensive dataset."
3. **Comparative Performance of Models** The study evaluated five machine learning models. While most showed improvements when macroeconomic variables were added, tree-based models utilized the data most effectively.
4. **Feature Importance Validation** An analysis of feature importance confirmed that the models identified the macroeconomic variables as significant predictors of demand.
5. **Generalizability of Findings** The study serves to validate and generalize findings from a previous 2023 study by Haque, which focused solely on LSTM models.

Product age based demand forecast model for fashion retail

The primary difficulty lies in predicting sales for products that are new to the season, have short life cycles, and require ordering decisions 6–12 months in advance. The authors introduce a novel "age-based" prediction model that utilizes machine learning (XGBoost) and Deep Learning (ANN) techniques.

Unlike traditional statistical models that rely on historical sales of specific items, this approach models demand as a function of product age (weeks since launch), attributes (specifically using RGB color feature engineering), price, and store clustering. The framework extends beyond forecasting to include specific recommendations for product launch timing and assortment optimization using Integer Linear Programming. When applied to a real-world use case of a multinational retailer with over 300 stores, the framework demonstrated a significantly lower forecast error than baseline models and projected a 41% revenue uplift compared to the retailer's existing plan.

Key points:

1. **The Challenge of "Cold Start" Forecasting in Fashion** The paper highlights that traditional forecasting methods fail in fashion because most items are new and lack historical data.
2. **The "Product Age" Methodology** The core innovation is shifting the time axis of sales data to align with the product's life cycle (Age) rather than calendar dates. This allows the model to learn sales patterns relative to how long an item has been on the floor.
3. **Novel Feature Engineering (RGB Vectors)** Instead of standard one-hot encoding for colors, which treats all colors as equidistant/unrelated, the authors encoded colors using Red-Green-Blue (RGB) intensities to capture visual similarities.
4. **Integrated Supply Chain Optimization** The model is not limited to prediction; it feeds into a larger framework for determining the optimal launch week and the ideal store assortment (quantity and mix of items) to maximize revenue under constraints.

Fashion Retail: Forecasting Demand for New Items

This paper addresses a critical challenge in the fashion retail industry: forecasting demand for new, "cold start" items that lack historical sales data. Traditional time-series models fail in this context because they rely on past performance, while existing clustering methods often fail to account for the complex interaction of pricing, visibility, and merchandising.

The authors propose a machine learning approach that infers demand based on clothing attributes (style, material, color) and merchandising factors (discounts, visibility). They experiment with various architectures, including tree-based models (XGBoost, Random Forest) and deep learning models (MLP, LSTM). The study concludes that gradient-boosted tree models, particularly XGBoost, currently outperform deep learning architectures for this specific problem. The successful deployment of these models at Myntra resulted in significant improvements in sell-through rates and inventory planning.

Key points:

1. **The "Cold Start" Problem in Fashion** Traditional forecasting methods are inadequate for the fast-paced fashion industry, where trends are transient and new designs are constantly introduced without prior sales history.
2. **Limitations of Visual Clustering** The researchers found that grouping items solely by visual similarity or design attributes was insufficient for predicting sales behavior, as external factors often weigh more heavily on consumer decisions.
3. **Feature Engineering Strategy** The model relies on a combination of fashion attributes and engineered "merchandising factors" to predict demand. Crucially, features are transformed so the model does not need to memorize training data to predict future values.

4. **Performance of Tree-Based Models vs. Deep Learning** Despite the theoretical advantages of Long Short-Term Memory (LSTM) networks for sequence data, tree-based models provided superior results in this study.
5. **Business Impact and Deployment** The models were deployed in real-world settings for assortment planning and drop planning, showing tangible financial benefits.

Machine Learning Algorithms with Intermittent Demand Forecasting: An Application in Retail Apparel with Plenty of Predictors

This study addresses the complexities of demand forecasting in the apparel retail industry, specifically focusing on "intermittent demand"—a pattern characterized by periods of zero sales followed by variable demand. The authors, Güven, Uygun, and Şimşir, compare two non-parametric machine learning algorithms: Random Forest (RF) and K-Nearest Neighbor (KNN).

To conduct this analysis, the researchers utilized four distinct datasets covering 212 weeks of sales between 2014 and 2018. A distinguishing feature of this study is the inclusion of 28 distinct predictor variables, ranging from product-specific attributes (color, price) to macroeconomic indicators (inflation, gold prices, tourism). The study concludes that Random Forest significantly outperforms KNN in handling the high standard deviation and intermittent nature of retail apparel data. Additionally, the study utilizes the RF algorithm to rank the importance of the predictor variables, providing insights into which factors most heavily influence forecast accuracy.

Key points:

1. **The Challenge of Intermittent Demand in Retail** The authors argue that traditional forecasting methods struggle with the specific nature of apparel sales, which often drop to zero due to seasonality or fashion trends.
2. **Methodology: Random Forest (RF) vs. K-Nearest Neighbor (KNN)**The study applied both algorithms to four datasets (CNS, MLN, RTK, TNT) using K-fold cross-validation to ensure reliability.
3. **Superior Performance of Random Forest** The results consistently favored RF across all datasets. While KNN managed to identify some zero-sales patterns, it struggled with high-volume deviations.
4. **Identification of Variable Importance** A key utility of the RF method was its ability to rank the 28 predictors based on their impact on forecasting error.
5. **Handling High Variance and Outliers** Both models faced challenges with datasets containing high standard deviations, particularly the RTK dataset.

Found Datasets

I found the following datasetes that can be used during Master Thesis research:

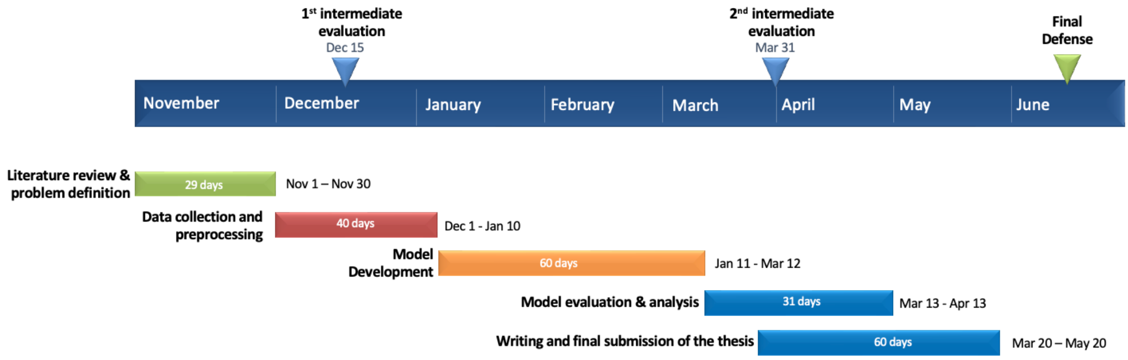
1. “Adidas vs Nike”. Adidas vs Nike is a constant debate in the sports industry. This dataset consists of the product information of these two huge companies with significant information.
2. “Retail Sales Index internet sales”. Dataset shows internet sales in Great Britain by store type, month and year.

3. "H&M Personalized Fashion Recommendations". The purchase history of customers across time, along with supporting metadata is presented.

All this data is real, and it is needed to be analyzed more precise to understand wheter this data is suitable for the reasearch.

Further Development Plan

Now, everything is going according to the plan. The next main step is to collect more data if it is possible and preprocess it. All further steps are shown in the picture.



References

1. C Giri, Y Chen (2022). Deep Learning for Demand Forecasting in the Fashion and Apparel Retail Industry. https://doi.org/10.3390/forecast4020031
2. IF Chen, CJ Lu (2021). Demand Forecasting for Multichannel Fashion Retailers by Integrating Clustering and Machine Learning Algorithms. https://doi.org/10.3390/pr9091578
3. PK Singh, Y Gupta, N Jha, A Rajan (2019). Fashion Retail: Forecasting Demand for New Items. https://doi.org/10.48550/arXiv.1907.01960
4. M Kunz, S Birr, M Raslan, L Ma (2023). Deep Learning Based Forecasting: A Case Study from the Online Fashion Industry. https://doi.org/10.48550/arXiv.2305.14406
5. S Anitha, R Neelakandan (2024). A Demand Forecasting Model Leveraging Machine Learning to Decode Customer Preferences for New Fashion Products. https://doi.org/10.1155/2024/8425058
6. A Mitra, A Jain, A Kishore, P Kumar (2022). A Comparative Study of Demand Forecasting Models for a Multi-Channel Retail Company: A Novel Hybrid Machine Learning Approach. https://doi.org/10.1007/s43069-022-00166-4
7. İ Güven, Ö Uygun, F Şimşir (2021). Machine Learning Algorithms with Intermittent Demand Forecasting: An Application in Retail Apparel with Plenty of Predictors. https://doi.org/10.32710/tekstilvekonfeksiyon.809867

8. RK Vashishtha, V Burman, R Kumar (2020). Product Age Based Demand Forecast Model for Fashion Retail. https://doi.org/10.48550/arXiv.2007.05278